

Core Tech Road Accident Dashboard

2021–2022 Casualty Analysis | Severity, Road, Vehicle & Conditions



Agenda

01

Dataset scope and measures

Metric definition and coverage

03

Severity distribution

Fatal, serious, and slight breakdown

05

Road characteristics

Type and surface conditions

07

Key insights

Pattern synthesis

02

Dashboard overview

Visual summary of key data

04

Vehicle impact analysis

Car vs non-car casualties

06

Context factors

Urban/rural and lighting

08

Recommendations

Next steps for intervention

Dataset Scope & Measures

Primary Metric

Number of casualties

Time Period

2021–2022 combined

Severity Levels

Fatal, Serious, Slight

Context Dimensions

- Vehicle group classification
- Road type and geometry
- Road surface condition
- Urban vs rural location
- Light condition at incident



Purpose: Fast comparison and prioritization of high-volume intervention areas

ACCIDENT DASHBOARD

Serious Cases

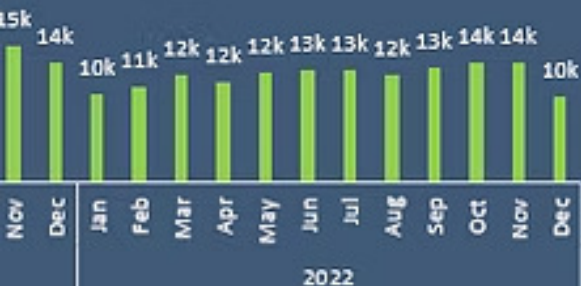
40740



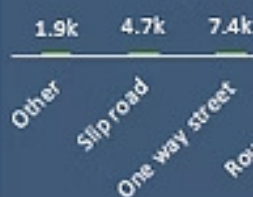
Slight Casualties

263279

Total



Casualties



Casualties by Light Condition



Dashboard Overview

Interactive summary of 417,882 casualties across multiple dimensions

Comprehensive view integrating severity, vehicle type, road conditions, and temporal factors

Severity Snapshot

417,882

Total Casualties

Combined 2021-2022

3,953

Fatal

Highest severity incidents

40,740

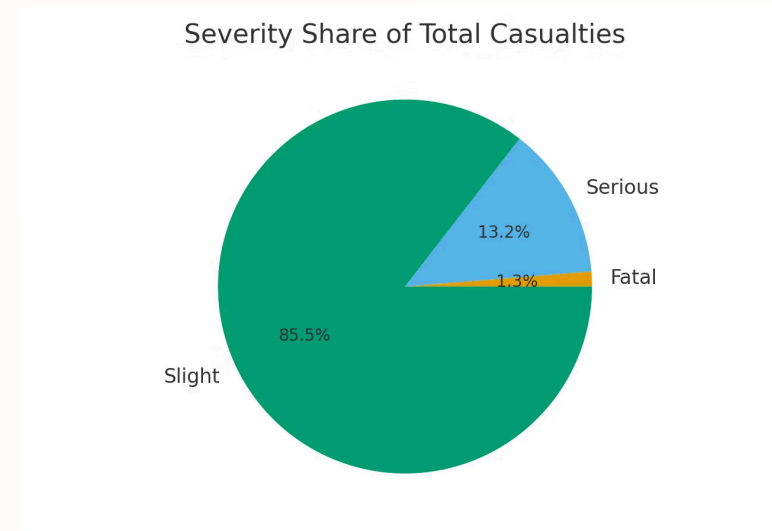
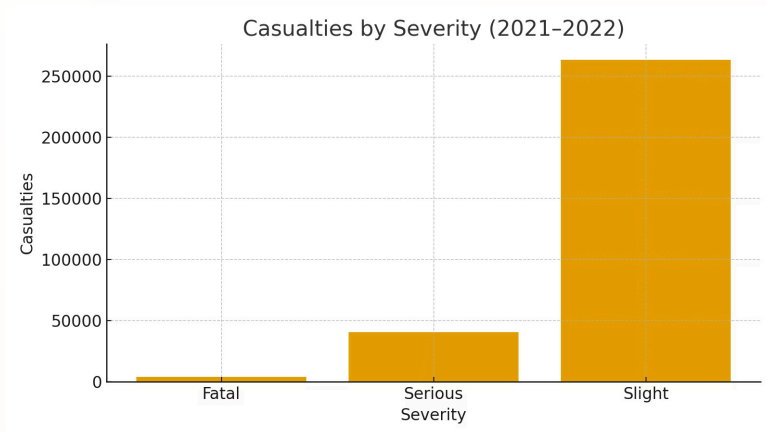
Serious

Significant injury cases

263,279

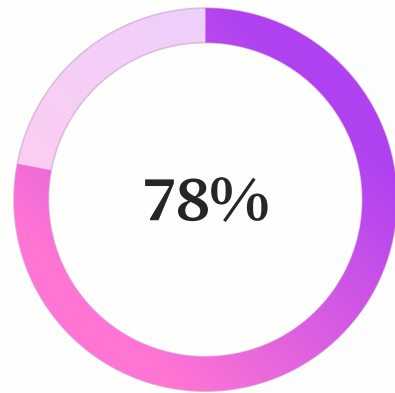
Slight

Minor injury incidents



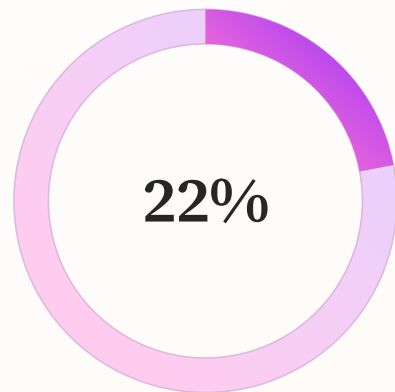
Slight injuries represent the dominant share, driving overall casualty burden

Vehicle Impact Analysis



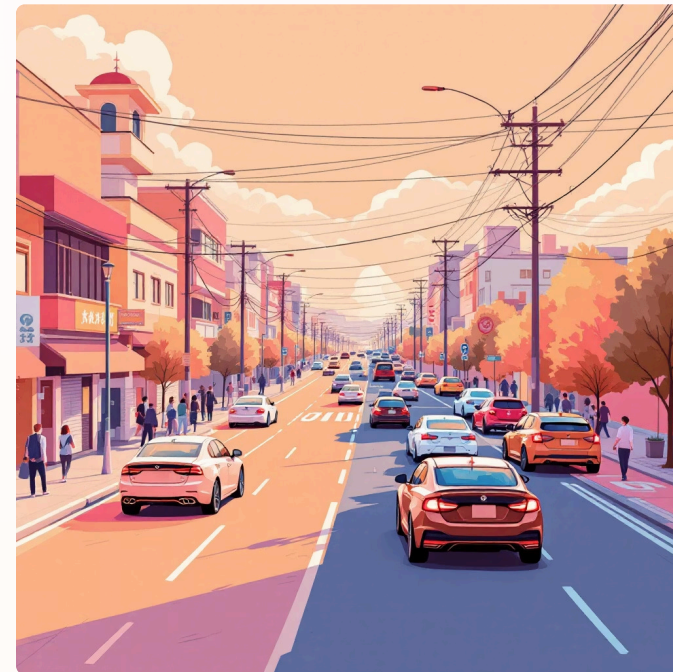
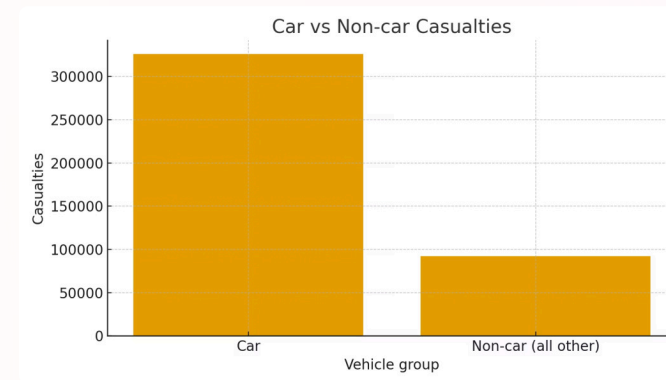
Car-Related

325,921 casualties



Other Vehicles

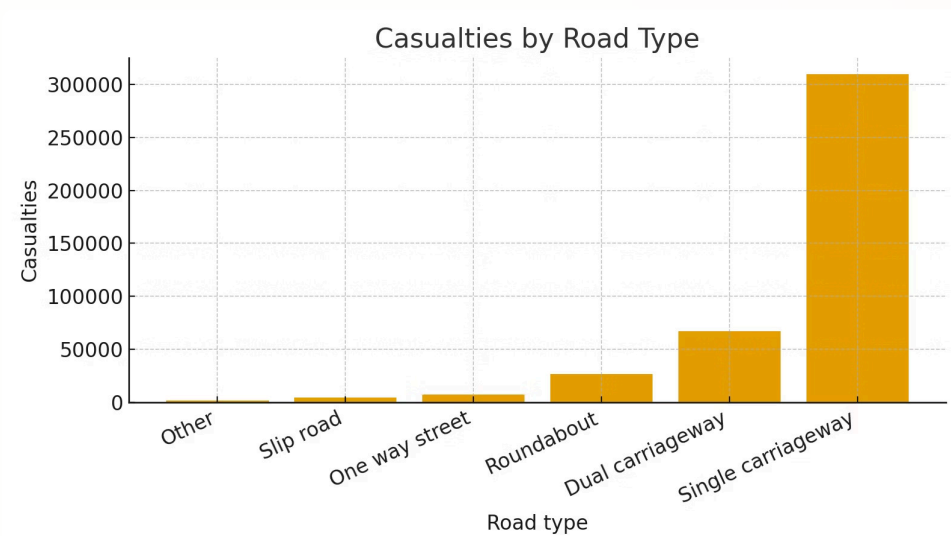
91,961 casualties



Key Interpretation

- High car volume reflects both exposure and behavioral factors
- Absolute reduction potential is greatest in private vehicle category
- Enforcement and public awareness can yield large-scale impact

Casualties by Road Type



Single Carriageways

Largest volume contributor

Dual Carriageways

Secondary volume level

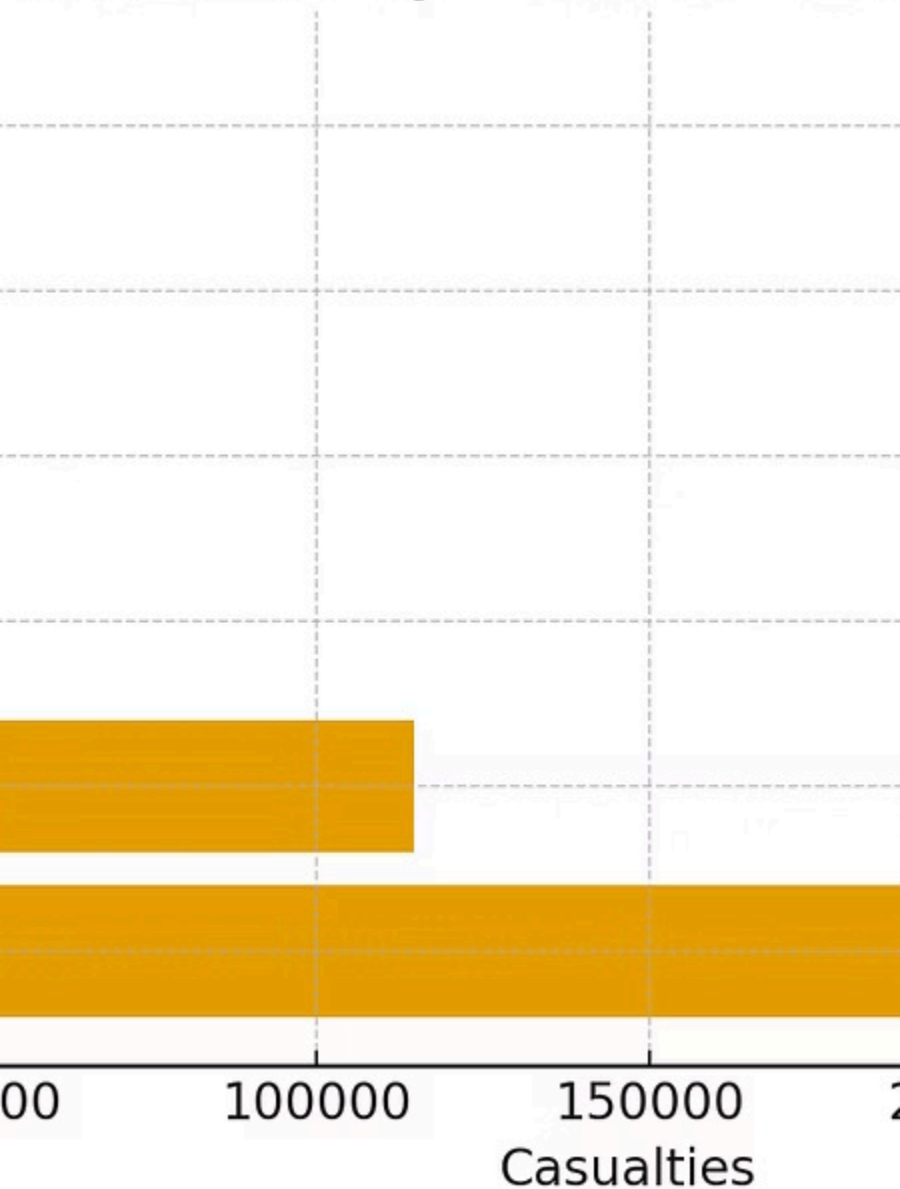
Roundabouts

Lower but notable share

Analysis Note: Volume reflects usage patterns. Use hotspot analysis to identify specific high-risk segments for targeted engineering improvements.



Casualties by Road Surface



Road Surface Conditions

Dry Conditions

Highest volume—likely reflects greater exposure rather than increased risk per trip

Wet/Damp Surface

Substantial second-largest share—elevated friction risk during precipitation

Frost/Snow/Flood

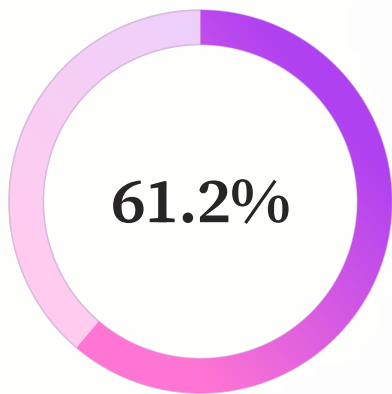
Smaller volume but potentially higher per-trip risk—seasonal hazard focus needed

Intervention Opportunities

Improved drainage systems, anti-skid surfacing, seasonal safety messaging campaigns

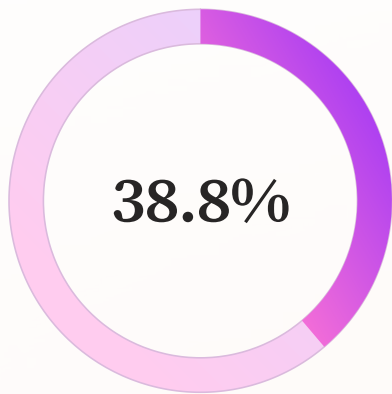
Location & Light Condition

Urban vs Rural



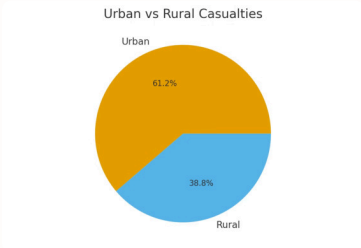
Urban Areas

255,900 casualties

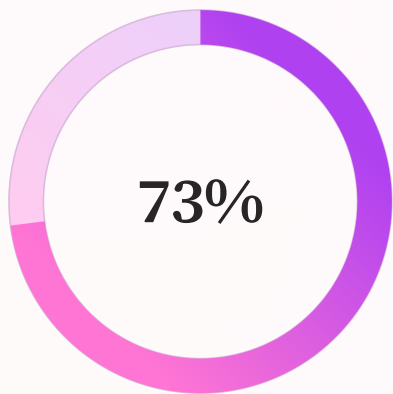


Rural Areas

162,000 casualties

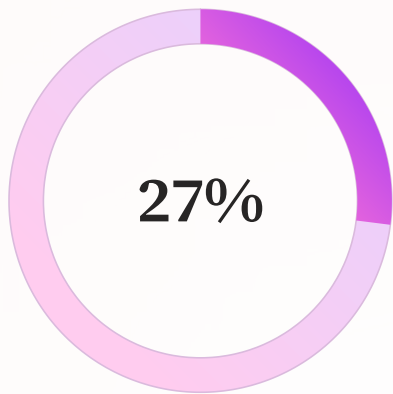


Light Condition



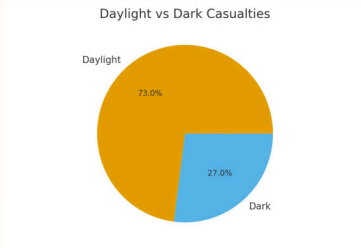
Daylight

305,000 casualties



Dark

112,900 casualties



📌 **Strategic Implication:** Combine urban design safety measures with time-specific enforcement targeting night-time risk behaviors



Key Insights



Slight Injuries Dominate

Highest volume category driving overall casualty burden



Car Incidents Lead

78% of casualties—largest intervention opportunity



Single Carriageways

Largest road-type contributor by volume



Dry & Daylight

Lead in counts—exposure-driven patterns



Urban Dominance

Higher casualty volumes than rural areas



Priority Strategy

Target high-volume categories for maximum impact